Batch #3 is here, limited kits

Got it!

BaBot

Thank you for being a BaBot owner. This manual will guide you step by step through the assembly process, from identifying each part to getting your robot up and running.

If you have any questions or run into any issues along the way, don't hesitate to leave a comment at the end of the page. Your feedback and questions are always welcome.

Let's get started — the fully assembled BaBot is just a few steps away!

Step 1

Begin by assembling the arms, which consist of six parts in total.

- 2 white plastic parts
- 1 metal ball
- 2 screws
- 1 set screw

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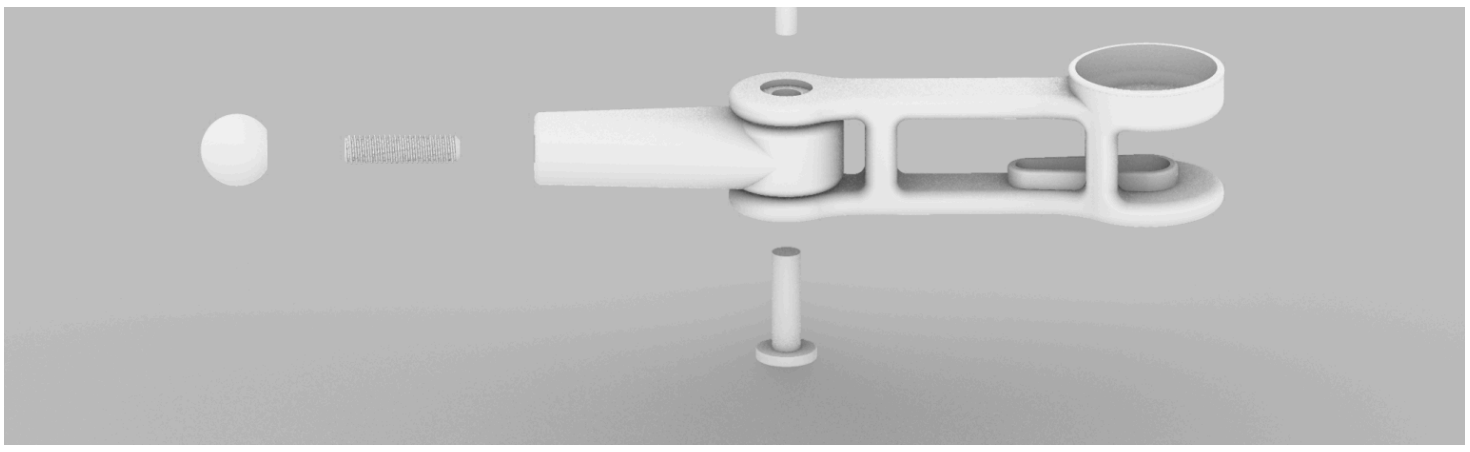
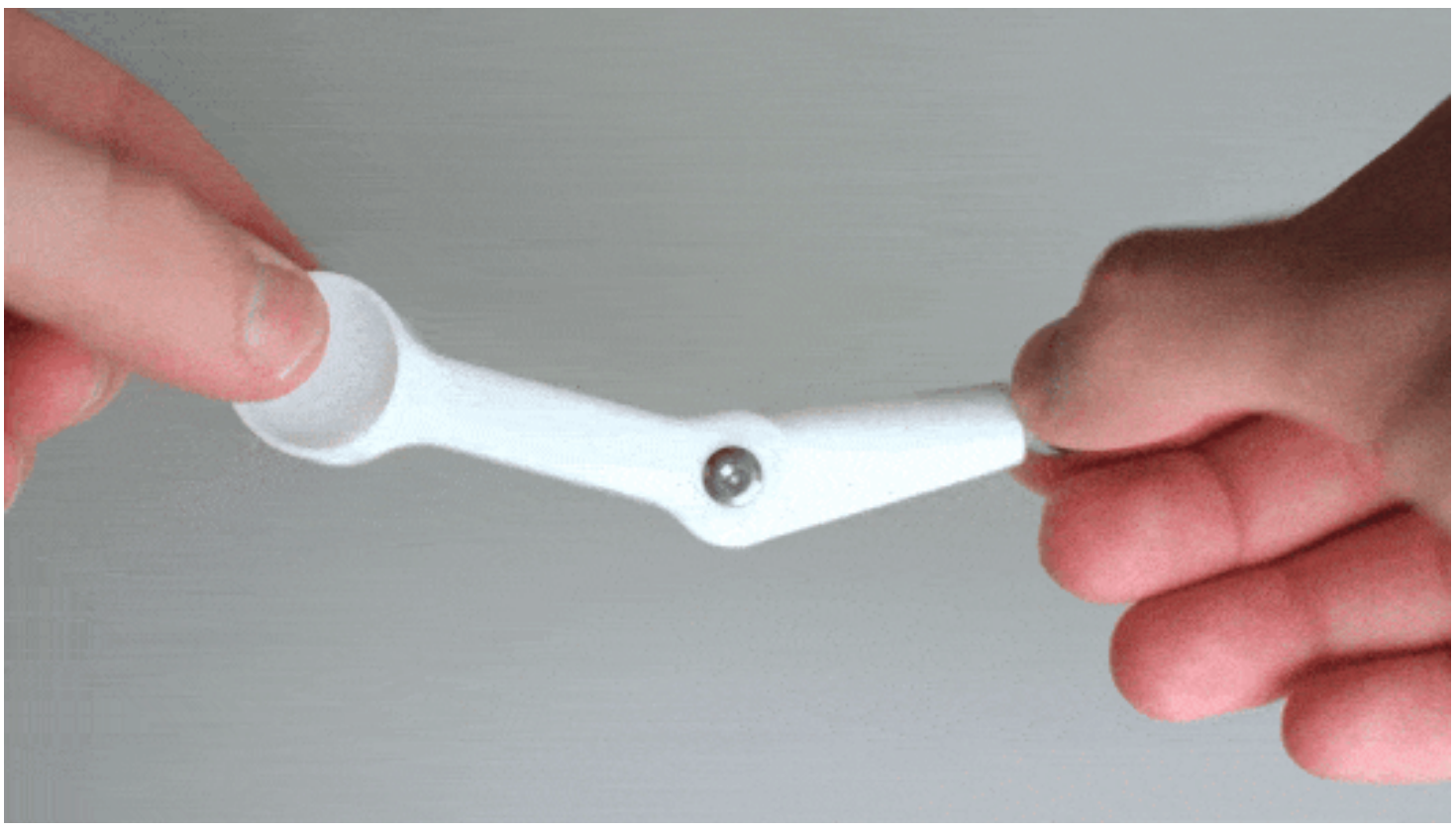


When securing the parts with screws, be careful **not to tighten them too much**. The joint should remain secure but still rotate easily with very low friction.



Step 2

Use the provided tool to unscrew the electronic circuit from the base by removing the 3 small screws. Once the screws are taken out, you'll have access

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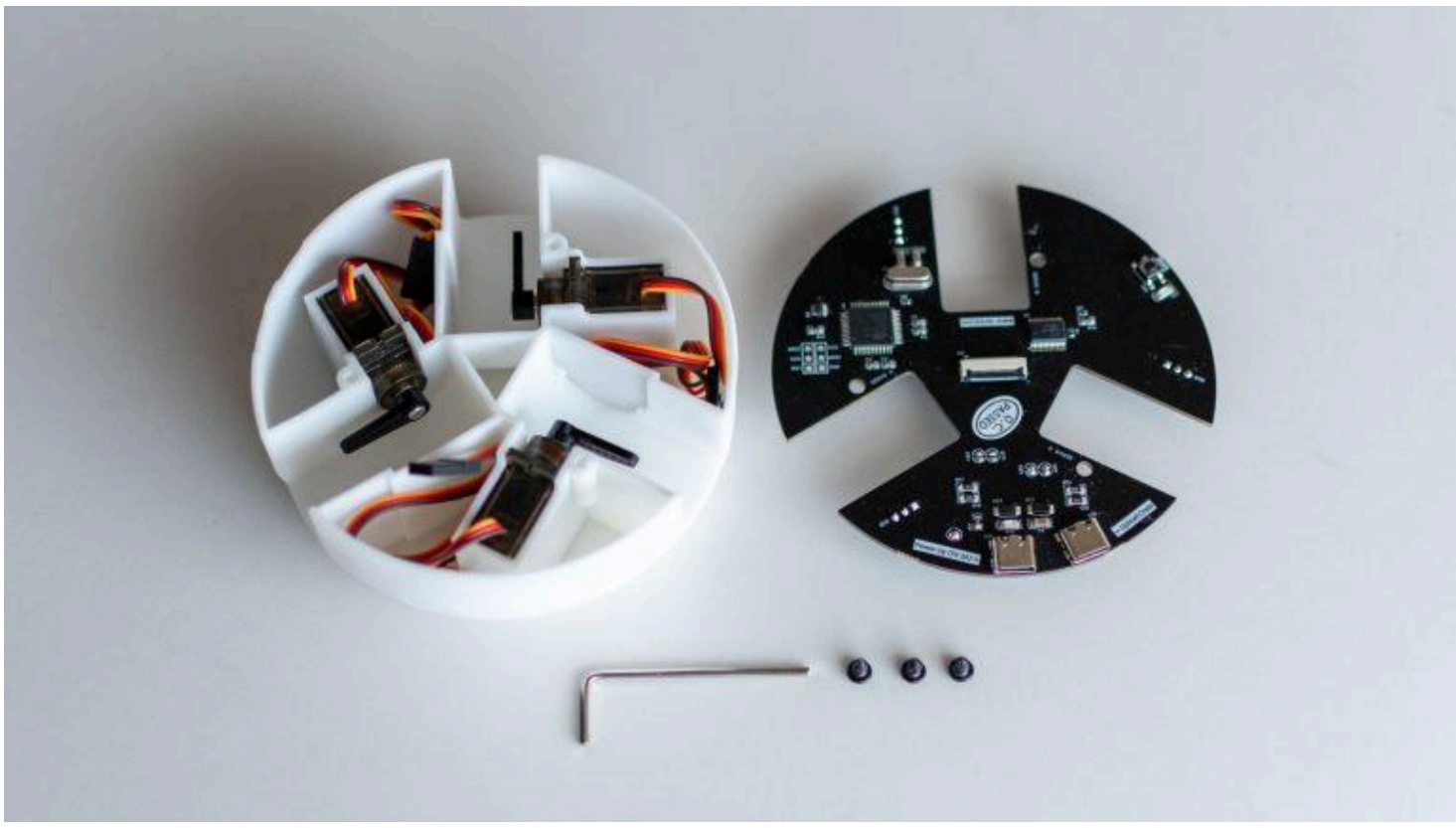
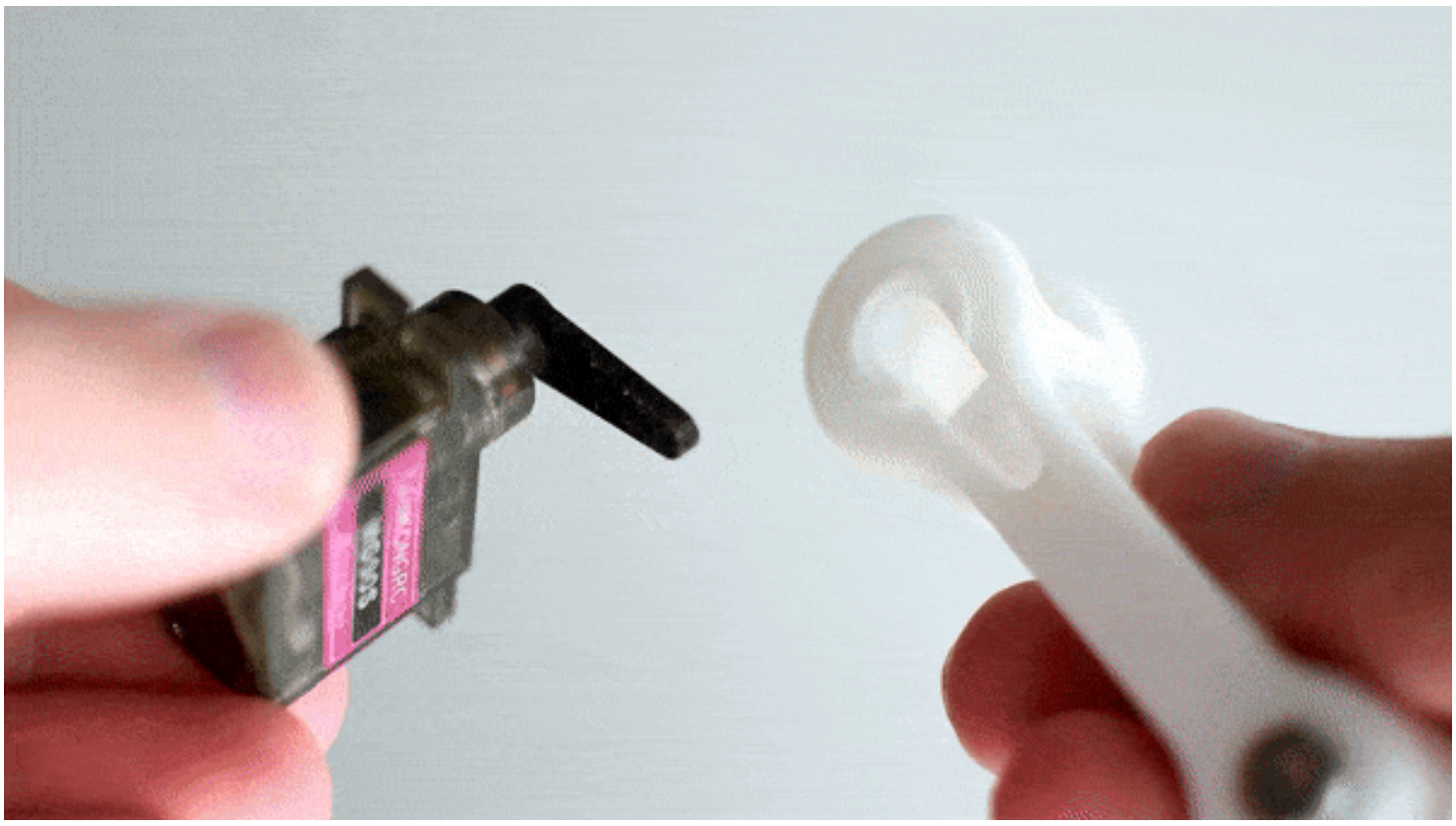


Step 3

Fit the arms into the motor arm as shown in the GIF below. Start by inserting the tip of the motor arm (black part) into the hole, then rotate and push the rest of the motor into place.

Repeat this process for each motor.



Step 4

Place the motors and their respective arms back into the base. Make sure to push each motor completely down into its position, ensuring a secure fit.

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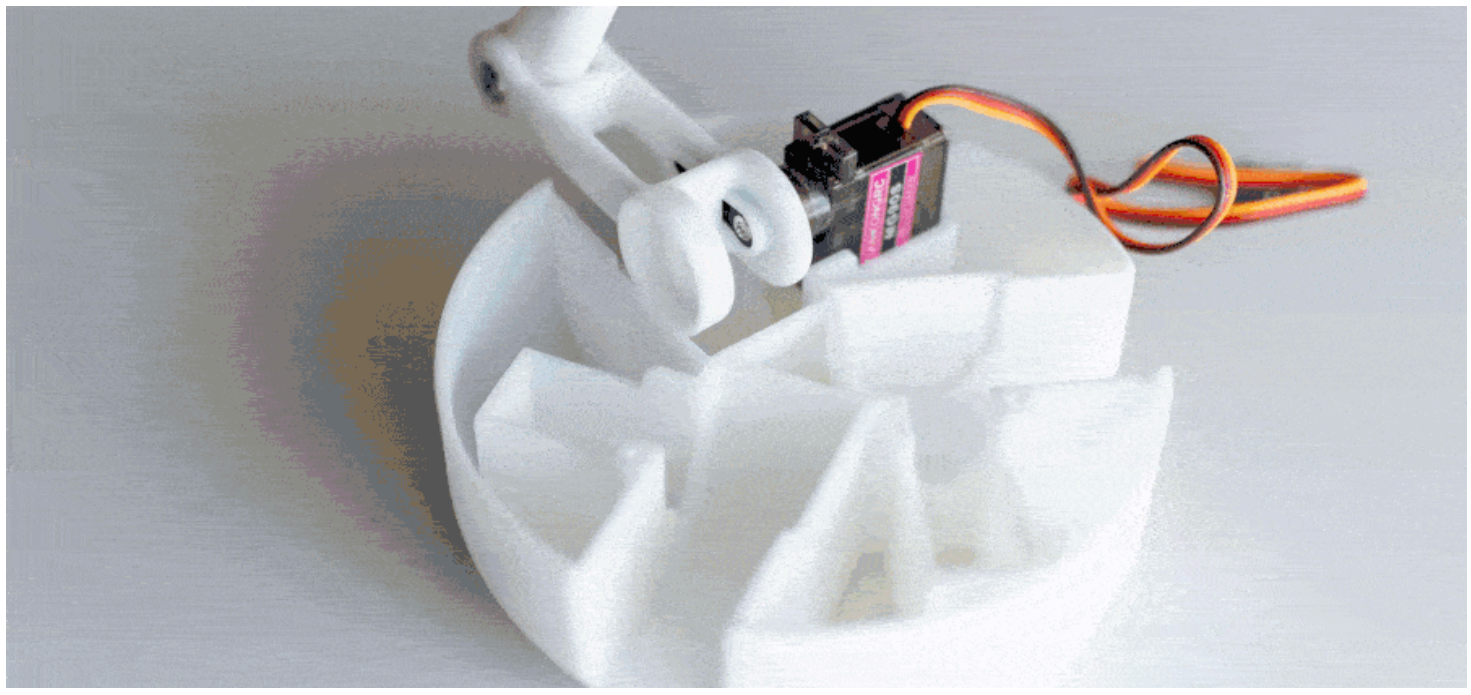
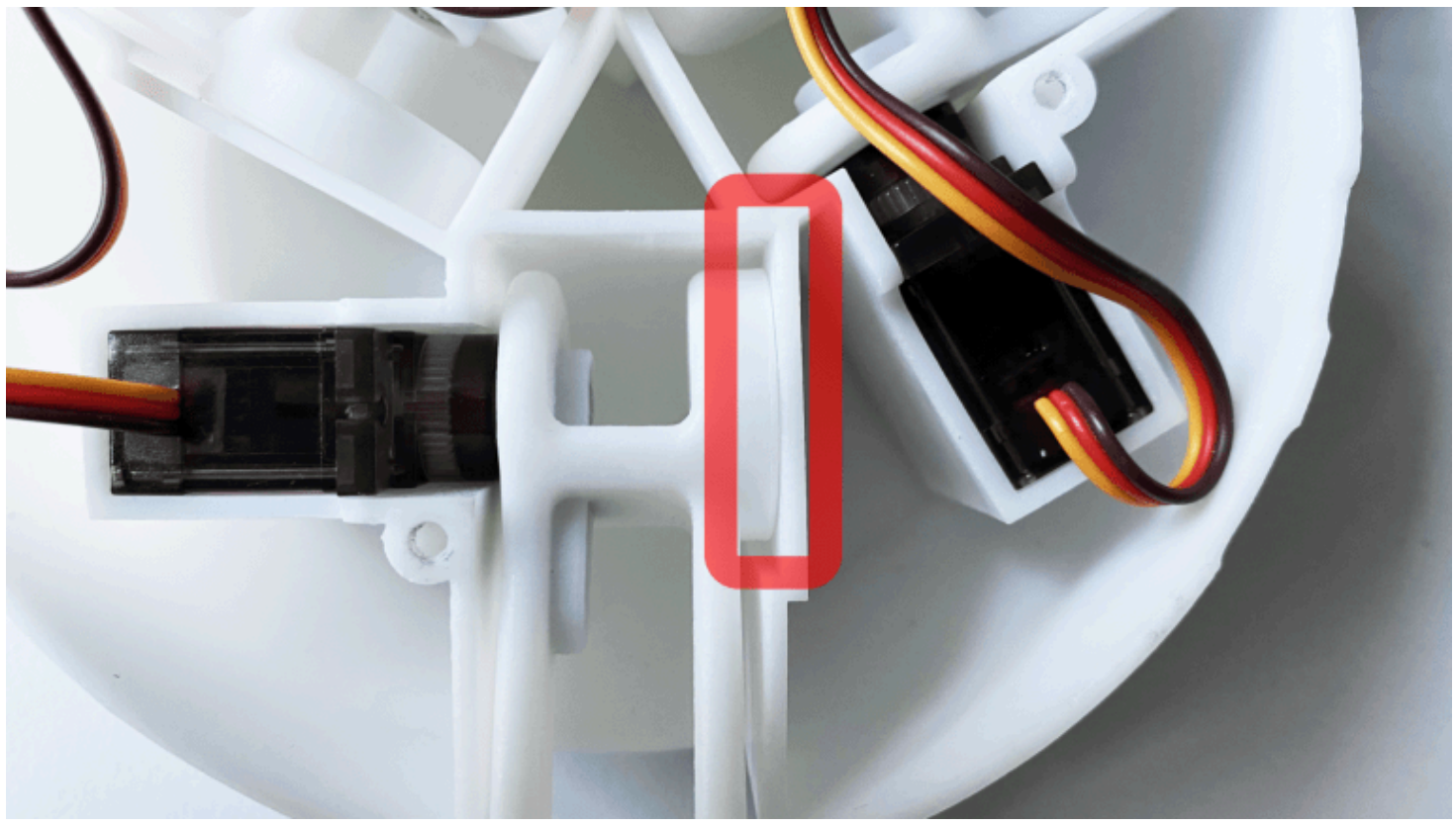
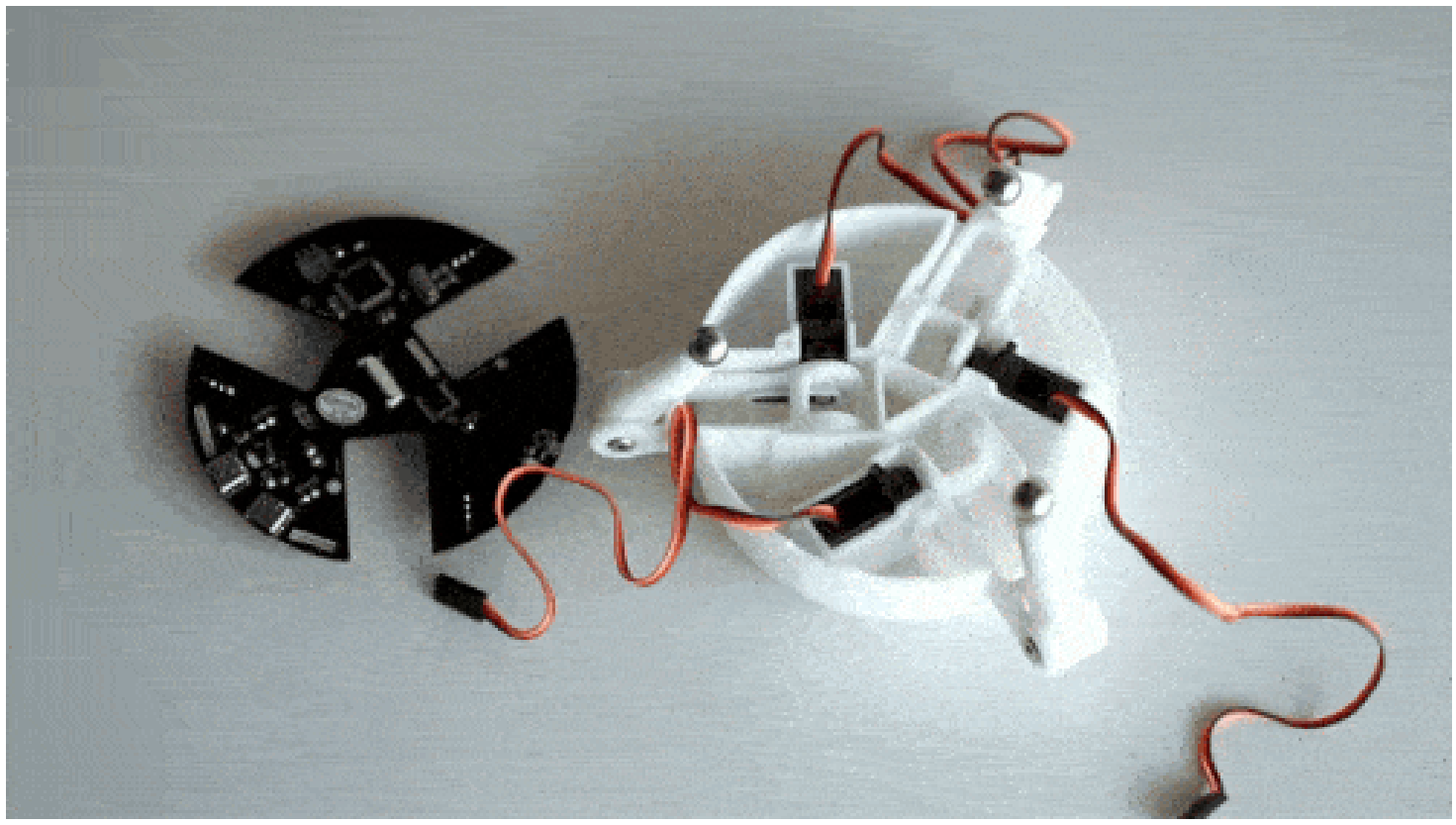


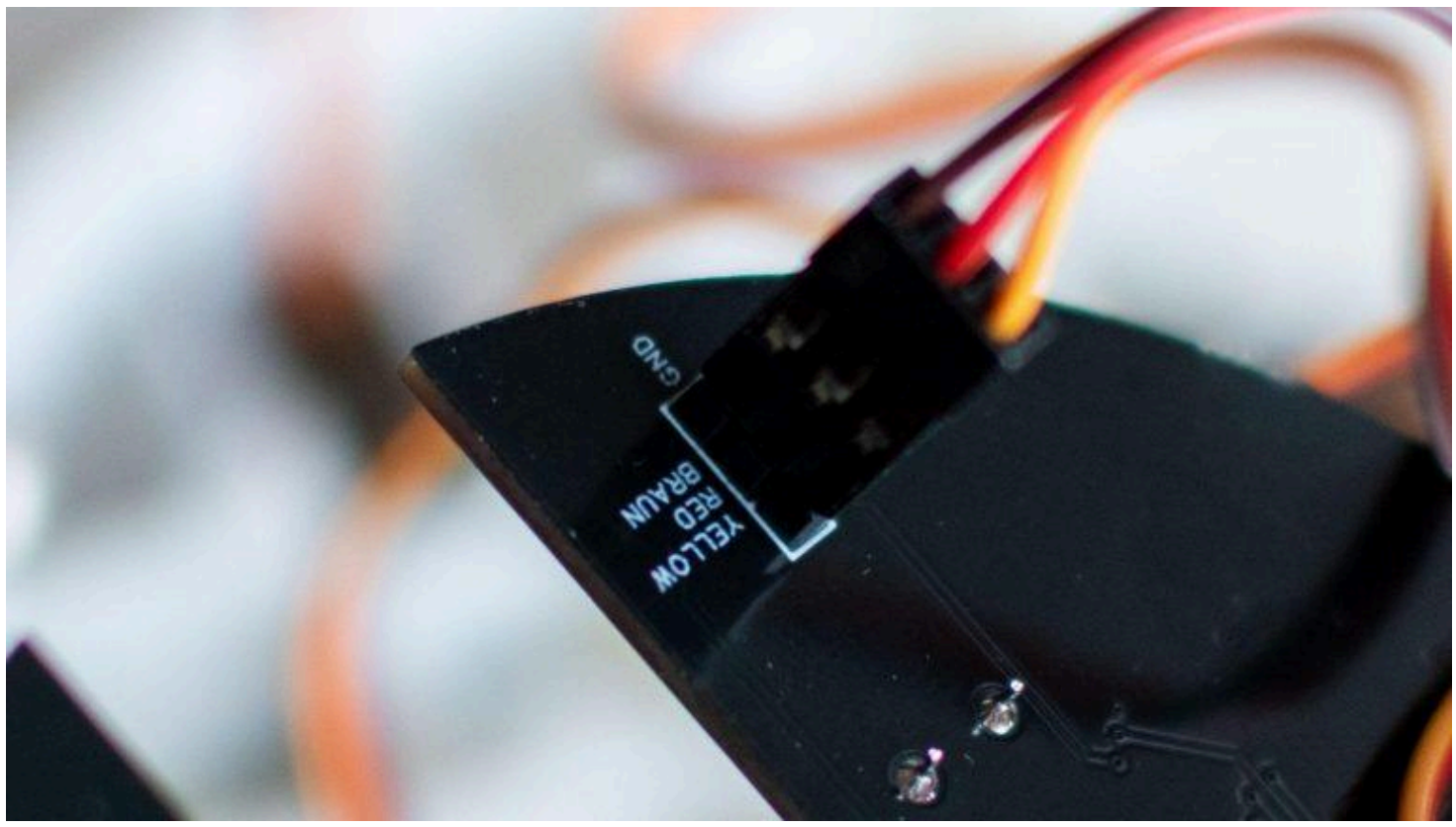
Make sure this part of the arm is not rubbing against the base. If it is, the motor may have difficulty rotating properly and this could cause issues. In that case, you can use a piece of sandpaper to remove a small amount of material. In the end, there should be a small gap between the arm and the base (see red rectangle below).



Step 5

Connect the motor cables to the circuit. Make sure to plug each connector in with the correct orientation, as reversing them could prevent the motors from working properly.





Place the circuit back into the base. Carefully bend the motor cables into the gaps around the motors so they don't interfere with the placement. The circuit should sit perfectly flat once everything is in position. Make sure to **align the two USB-C ports with the two small notches** on the base. Once aligned, screw back the three small screws to secure the circuit in place.

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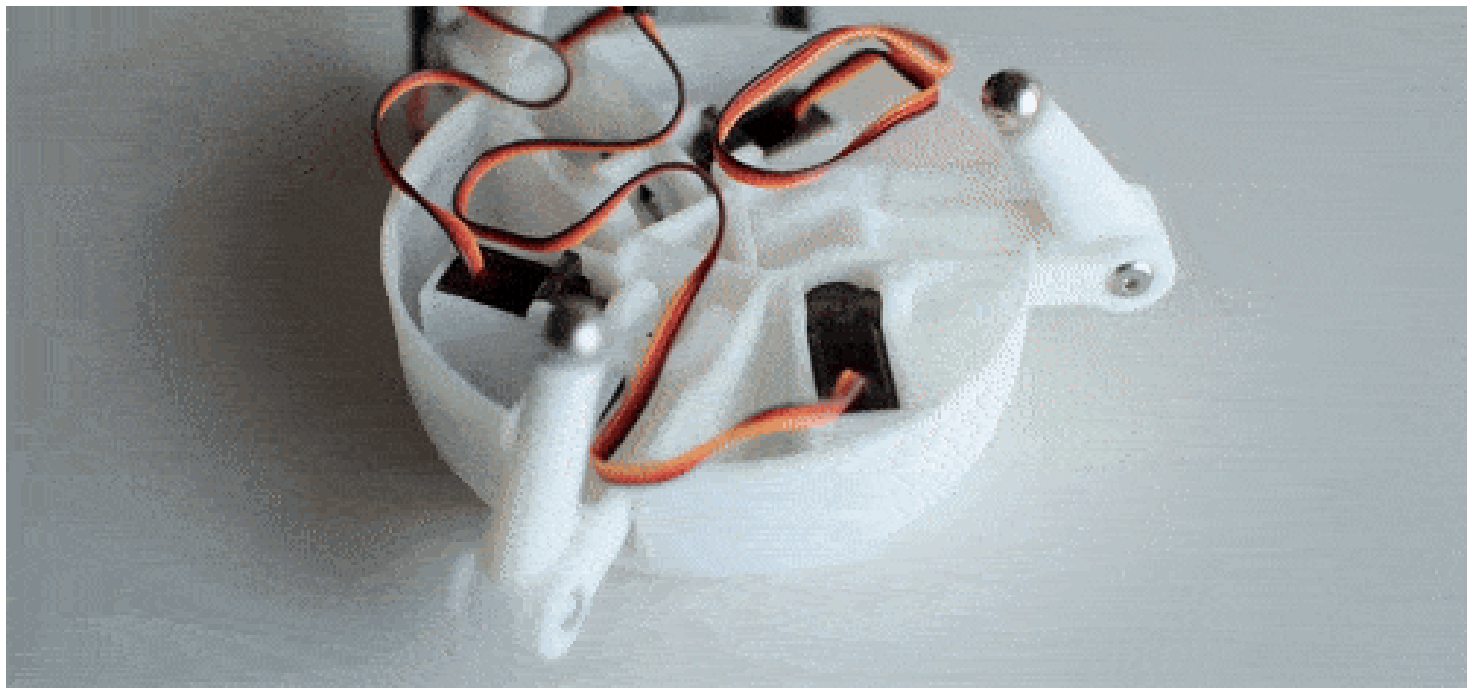
Got it!



Step 6

In this step, you will calibrate each motor so the BaBot plate is perfectly horizontal once installed.

Follow the process below and watch the video for a visual guide.

1. Connect the PCB “Power” port to your computer or to any USB-C phone charger to power up BaBot.
2. Triple press the button.
   The servos will move and servo A will position slightly lower than the other two.
3. Press the button repeatedly.
   Each press moves the arm down by 1°.
   Continue until the arm is as horizontal as possible.
4. When servo A is horizontal, long press the button for 2 seconds to confirm calibration.
5. Servo B will automatically move to its calibration position.
   Repeat the same process, press multiple times to adjust, then long press to confirm.
6. Repeat the same steps for servo C.

Calibration is now complete !

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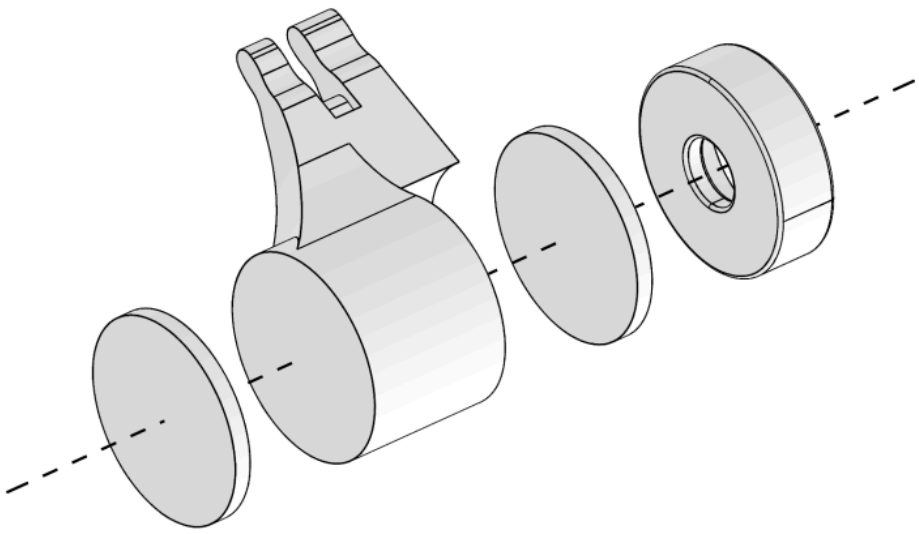
Got it!

0:00 / 0:53

Step 7

Now let's assemble the platform of BaBot. Start by taping the magnets onto the circuit holder, as shown in the images below.



Step 8

Start by taping two of the circuit holders onto the transparent plate (Make sure to remove the protective film from the plate beforehand), following the positioning shown below. Then, slide the circuit notches into the holders,

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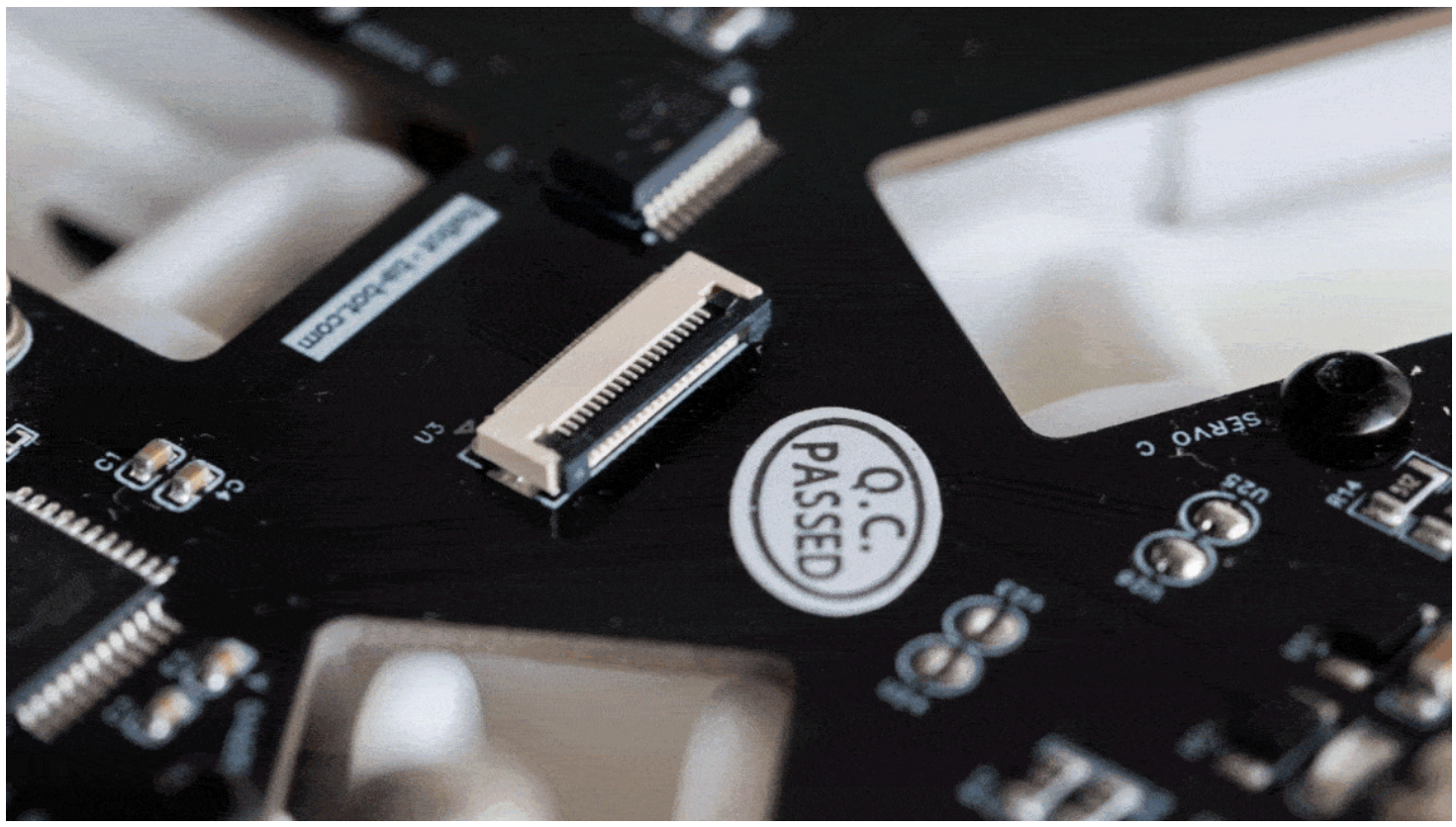
Got it!

Step 9

Connect the flat cable to the base circuit. First, open the small locking connector by gently lifting the lever. Carefully slide the flat cable into the connector, making sure it is fully inserted and properly aligned. Once in place, close the lever to lock the cable securely.



Step 10

Repeat the same process to connect the other end of the flat cable to the second circuit.

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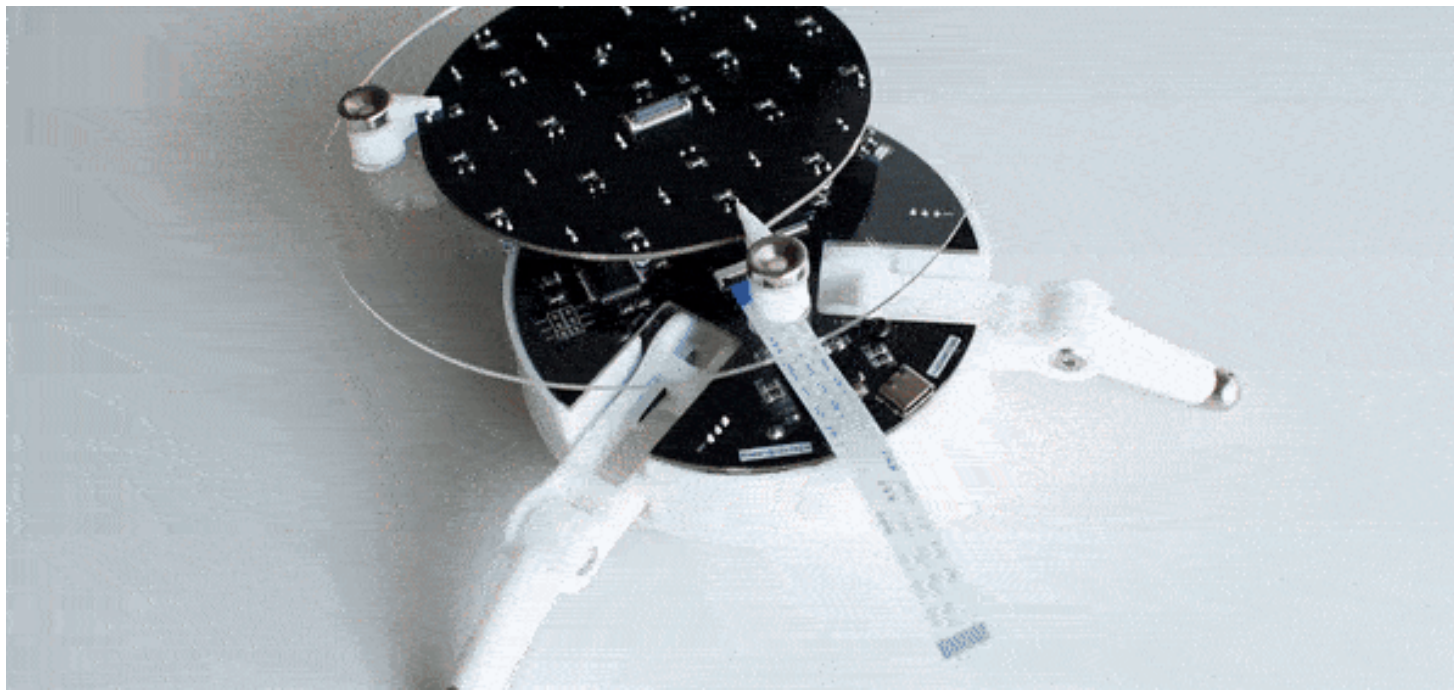
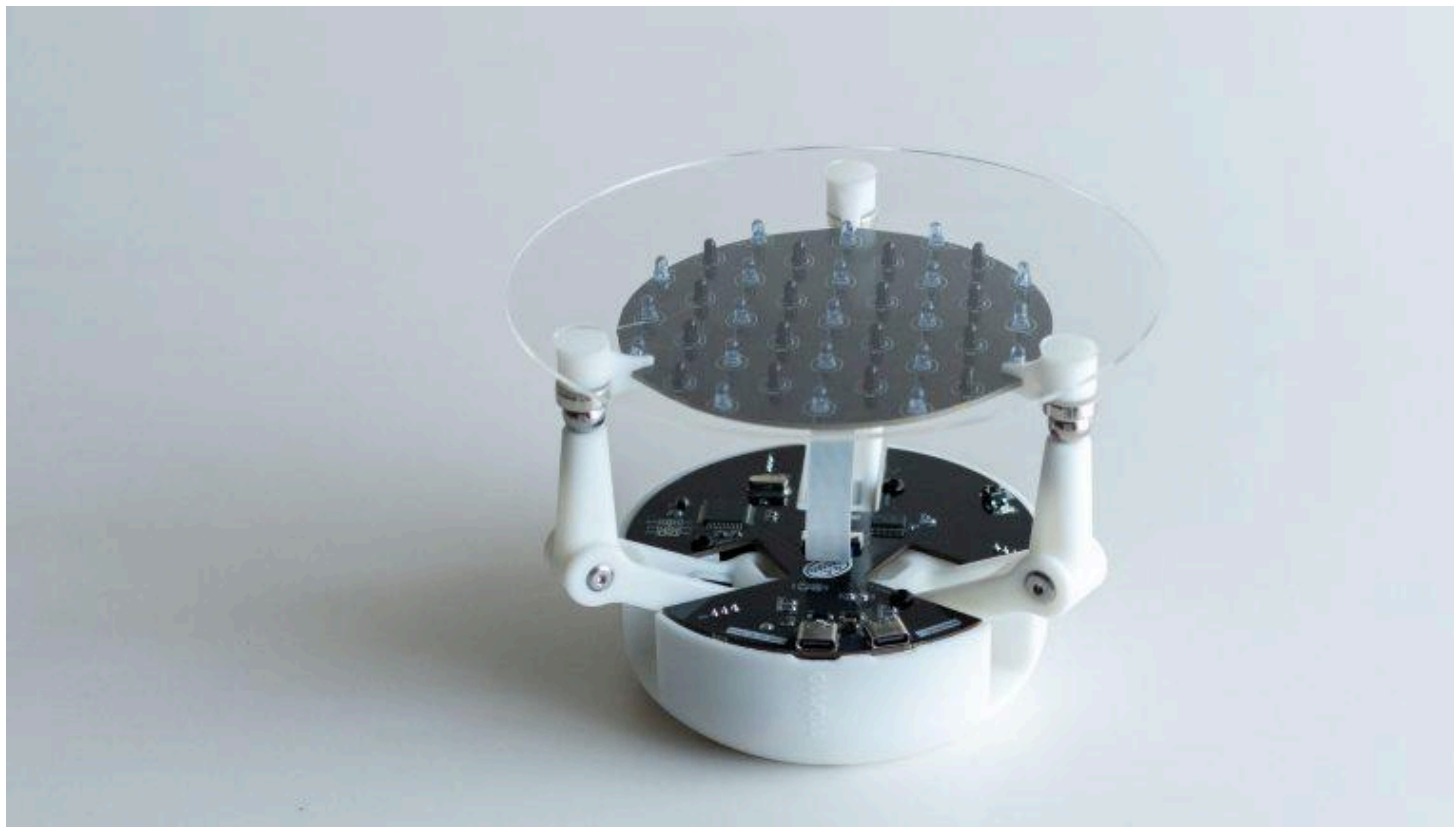


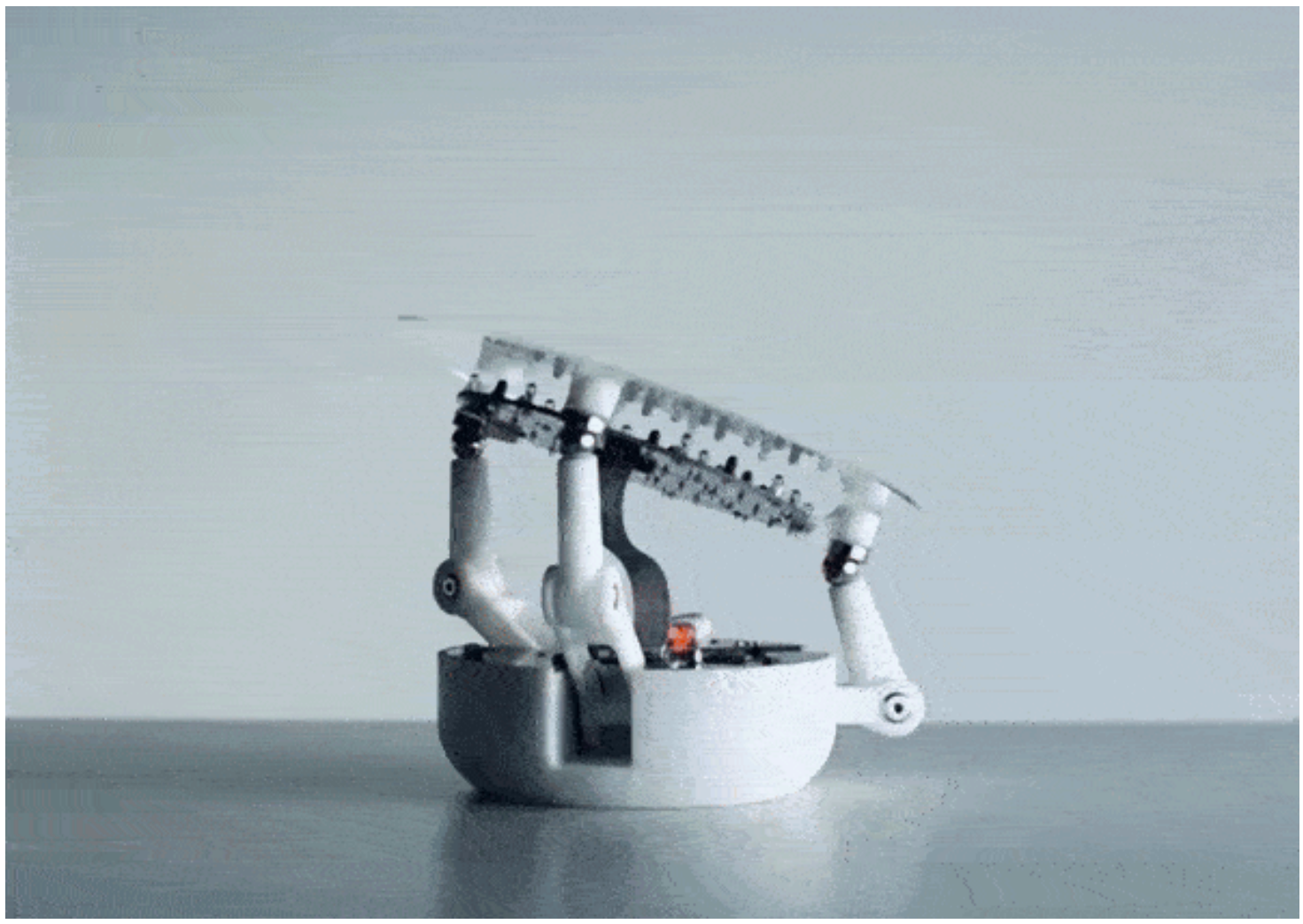
Step 11

Congratulations! You've successfully assembled BaBot.



How To Use It ?

To get started, plug the USB-C cable into the connector labeled "Power Up". The cable can be connected to either a computer or a standard phone charger. The platform should automatically move to a horizontal position, and the red LED next to the button will begin blinking. Once both conditions are met, BaBot is ready to receive a ball.

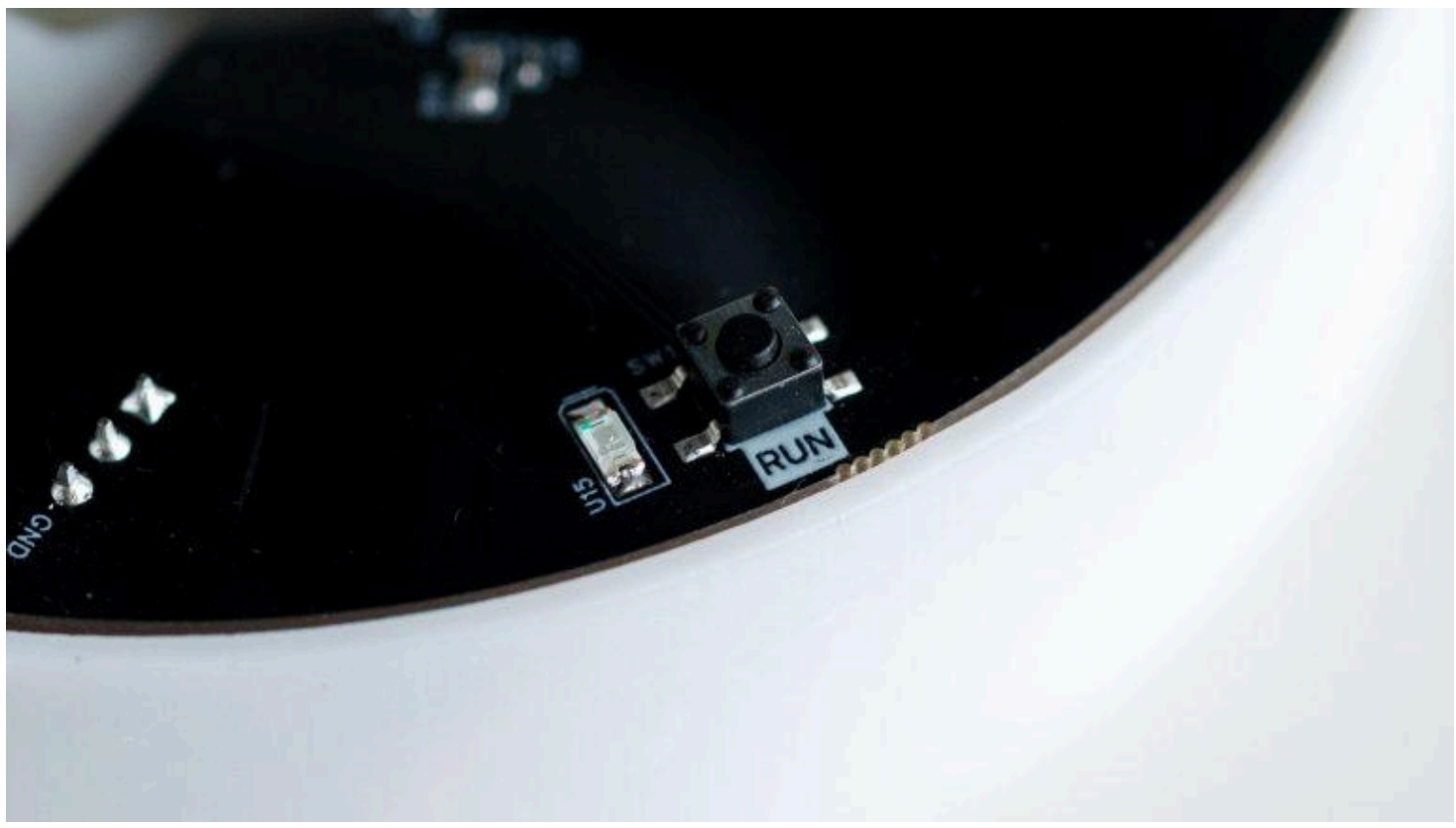


Simply place a ball on the platform to begin interacting with it. You can use BaBot for hands-on play or simply leave it on your desk to impress friends and colleagues.

LED Status Guide:

- Blinking – BaBot is on and ready to receive a ball
- Off – BaBot is powered off
- On (Solid Red) – BaBot has detected a ball

Pressing the button toggles BaBot on or off.



BaBot is highly sensitive to infrared light. It cannot be used outdoors or in very bright sunlight, as infrared interference from the sun can make the sensors completely unresponsive. For best performance, use BaBot indoors under controlled lighting conditions.

Troubleshooting

If you're experiencing problems, follow the steps below to identify and resolve common issues:

1. Check the Lighting Conditions
   Perform the first BaBot tests in a room with limited natural light. Excess ambient infrared (IR) light can interfere with the sensors. This is one of the most common issues users encounter.
2. Verify Smooth Motor Operation
   Ensure the motors are running smoothly.
   - If an arm appears stuck or doesn't rotate properly, return to Step 4 and confirm that no part of the arm is rubbing against the base.
   - Use sandpaper to carefully remove a small amount of material if needed, making sure there is a small gap between the arm and the base.
3. Re-upload the code
   If you place the ball on the plate and the LED continues blinking (indicating the ball is not detected), or if the ball is detected but fails to stabilize for more than 2–3 seconds, you may need to re-upload the code to the circuit:
   - Download and install the Arduino IDE on your computer
   - Download and open the BaBot code
   - Connect BaBot to your computer using the "Upload Code" USB port
   - (1) In the Arduino IDE, click on the Library Manager tab
   - (2) Type CD74HC4067 into the search box
   - (3) Install the Library
   - (4) Select "Arduino Leonardo" as the board type
   - (5) Click Upload and wait for the (6) "Done uploading" message (this should not take more than 20 seconds)
   - Once the upload is complete, disconnect the USB cable and reconnect it to the Power-Up port before placing a ball on the plate to test it.

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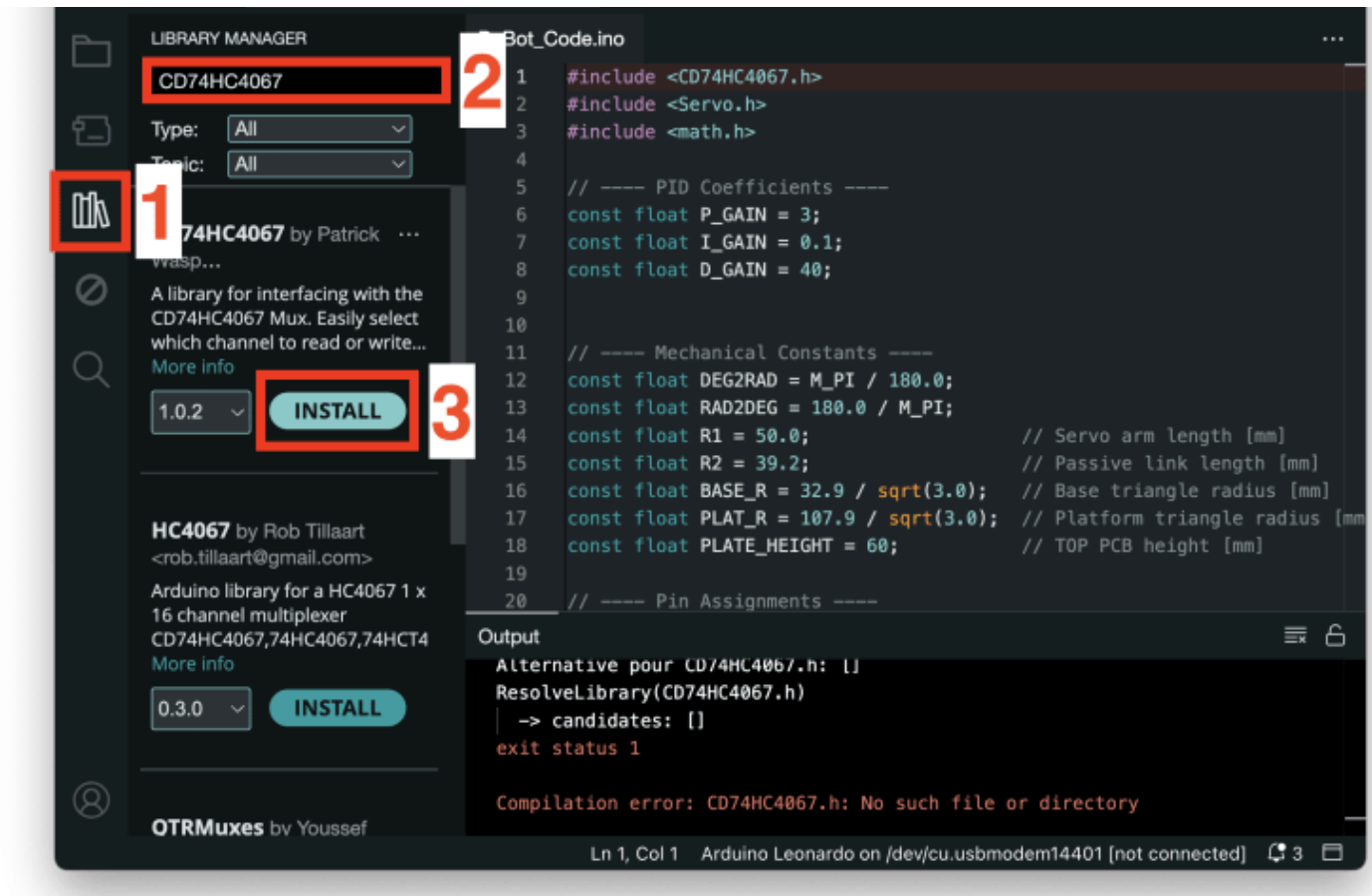

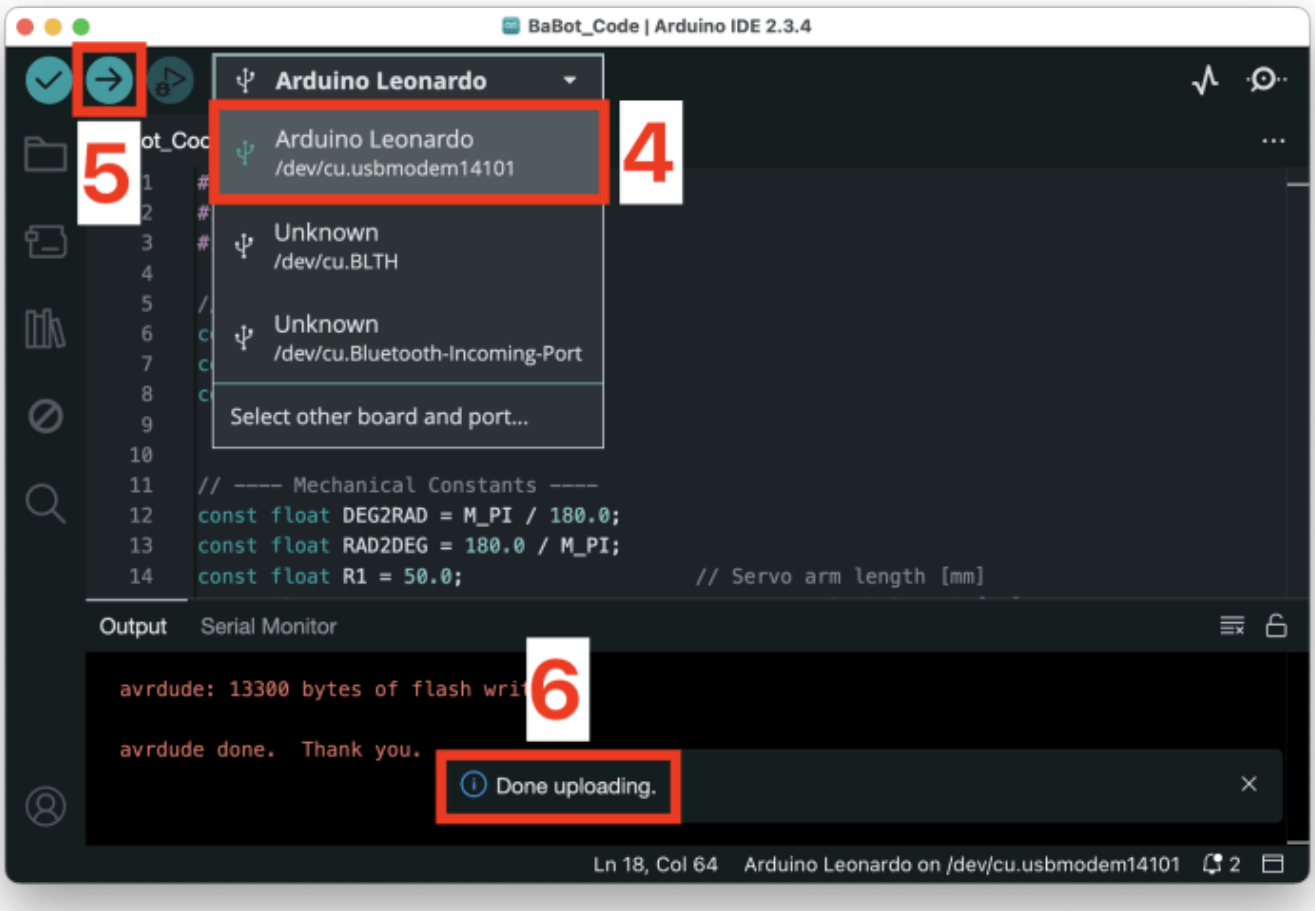

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Comment *